

SAFETY QUIZ:CHE382-Chemical Engineering Laboratory II

Please answer the following questions testing your knowledge from 6 selected chapters of the JT Baker Hazardous Chemical Safety document; Toxic Chemicals, Flammables, Corrosive Chemicals, Reactive Chemicals, Insidious Hazards and Compressed Gases. Be brief but concise in your answers. Open book/open notes and team discussion is allowed but answers must be in your own words. (5 points per question)

1. What is the general definition of a toxic chemical?

2. Name the general types of damage that a toxic chemical can cause:

3. Name two parameters that can effect toxicity:

4. What is the DOT (Department of Transportation) definition of a nonflammable compressed gas?

5. a) How does one determine how much material is remaining in a non-liquified gas cylinder?

b) On page 13-10, which pressure gauge on the two-stage regulator will give you the cylinder pressure reading?

University of Illinois at Chicago
Department of Chemical Engineering

6. Other than a single compressed gas, what else can be stored inside a compressed gas cylinder?

7. Why do insidious hazards represent a problem to workers?

8. What are the four general categories of insidious hazards?

9. Name 2 control measures that can be implemented to reduce insidious hazards in the laboratory.

10. What is the definition of a reactive chemical?

11. Name the two groups and their sub-headings used to categorize reactive chemicals:

University of Illinois at Chicago
Department of Chemical Engineering

12. List two precautionary measures for handling and/or storing reactive chemicals.

13. How are corrosive chemicals generally defined?

14. How do corrosive chemicals act on body tissues?

15. True or False: Concentrated alkaline solutions have a more corrosive effect on tissue than most strong inorganic acids.

16. Name the six categories of common corrosive liquids.

17. What category of corrosive solids presents the greatest hazard because of their frequent use in industry?

18. Name the three components that make up the FIRE TRIANGLE:

University of Illinois at Chicago
Department of Chemical Engineering

19. What category of flammables is the most common source of fires in industry?

20. List two common safety problems and two safety control measures for flammable solvents.

BONUS QUESTION:

Where is the eyeglass cleaning station located in the Unit Operations Laboratory (Room 120)?